MATH 345 Skeleton Notes
Unit 2: Anatomy of a linear map
Section 2.5: Inverses and determinants

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Matrix invertibility and linear systems

In this section we finish the square-matrix version of the Unit 2 story. A square matrix is invertible exactly when it "does not forget any nonzero input direction" and "every output is reached once". The inverse matrix gives the inverse map when it exists. The determinant, a fundamental quantity in linear algebra that we only briefly discuss in this course, gives a single-number test for invertibility.

Definition: Invertibility of Matrices

An $n \times n$ matrix A is called nonsingular, or invertible, if there exists an $n \times n$ matrix B such that $AB = BA = I_n$. Such a B is called the inverse of A. If no such B exists, A is called singular or noninvertible.

Note

The inverse of a matrix mimics the reciprocal of a real number.

Activity: Testing whether two matrices are inverses (U2-LO5)

Let

$$A = \begin{bmatrix} 1 & 3 \\ 1 & 4 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 4 & -3 \\ -1 & 1 \end{bmatrix}.$$

Compute both AB and BA, and make a conclusion using the language of inverses.

$$AB = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I_2 \quad \text{and} \quad BA = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I_2$$

Therefore, B is the inverse of A.

Fact

The inverse of a matrix, if it exists, is unique. Therefore, we can write A^{-1} for the inverse of A.

Fact

For $n \times n$ matrices A and B, if $AB = I_n$, then $BA = I_n$.

Activity: Testing a matrix for invertibility (U2-LO5)

Does $A = \begin{bmatrix} 2 & 1 \\ 0 & 0 \end{bmatrix}$ have an inverse?

Let's suppose A had an inverse $B = \begin{bmatrix} x & y \\ z & w \end{bmatrix}$.

Then:

$$AB = I_2$$

$$\begin{bmatrix} 2 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x & y \\ z & w \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

This gives us the system of linear equations in four unknowns:

$$\begin{aligned} 2x + z &= 1 \\ 2y + w &= 0 \\ 0x + 0z &= 0 \\ 0y + 0w &= 1 \end{aligned}$$

The last equation states that $0 = 1$, which is impossible.

Therefore, A cannot have an inverse. This means A is singular.

Theorem

If both A and B are nonsingular $n \times n$ matrices, then the matrix AB is

nonsingular and its inverse is $(AB)^{-1} = B^{-1}A^{-1}$.

Why is this true?
It suffices to calculate that $(AB)(B^{-1}A^{-1}) = I_n$ and $(B^{-1}A^{-1})(AB) = I_n$. Firstly,

$$(AB)(B^{-1}A^{-1}) = A(BB^{-1})A^{-1} = AI_nA^{-1} = AA^{-1} = I_n.$$

Similarly,

$$(B^{-1}A^{-1})(AB) = B^{-1}(A^{-1}A)B = B^{-1}I_nB = B^{-1}B = I_n.$$

Fact: Follow Up Facts About Invertibility

- If $A_1, A_2, \dots, A_k$ are $n \times n$ invertible / nonsingular matrices, then $A_1A_2 \cdots A_k$ is invertible / nonsingular and

$$(A_1A_2 \cdots A_k)^{-1} = A_k^{-1} \cdots A_2^{-1}A_1^{-1}.$$

- If A is invertible/nonsingular, then A^{-1} is invertible/nonsingular and

$$(A^{-1})^{-1} = A.$$

- If A is invertible/nonsingular, then A^T is invertible/nonsingular and

$$(A^{-1})^T = (A^T)^{-1}.$$

Let $T = T_A : \mathbb{R}^n \to \mathbb{R}^n$ be a matrix transformation induced by the $n \times n$ matrix A. What does invertibility imply about the geometry of the transformation T_A?
Consider $T' = T_{A^{-1}}$. What happens when we compose them?

Activity: Composing a map and its inverse (U2-LO5)

Compute the composition of T and T' in both orders.

Let's examine the composition of T and $T^{'}$:

$$\begin{aligned} T \circ T^{'}(\mathbf{x}) &= T_A(T_{A^{-1}}(\mathbf{x})) \\ &= T_A(A^{-1}\mathbf{x}) \\ &= A(A^{-1}\mathbf{x}) \\ &= (AA^{-1})\mathbf{x} \\ &= I_n\mathbf{x} \\ &= \mathbf{x} \end{aligned}$$

Similarly, for the reverse composition:

$$\begin{aligned} T^{'} \circ T(\mathbf{x}) &= T_{A^{-1}}(T_A(\mathbf{x})) \\ &= T_{A^{-1}}(A\mathbf{x}) \\ &= A^{-1}(A\mathbf{x}) \\ &= (A^{-1}A)\mathbf{x} \\ &= I_n\mathbf{x} \\ &= \mathbf{x} \end{aligned}$$

Definition

Given a matrix transformation $T : \mathbb{R}^n \to \mathbb{R}^n$, if there exists a transformation $S : \mathbb{R}^n \to \mathbb{R}^n$ such that $S(T(\mathbf{x})) = T(S(\mathbf{x})) = \mathbf{x}$ for all $\mathbf{x} \in \mathbb{R}^n$, we say that T has an inverse.

Theorem

Let $T : \mathbb{R}^n \to \mathbb{R}^n$ denote the matrix transformation induced by an $n \times n$ matrix A. Then:

1. A is invertible if and only if T has an inverse.
2. In this case, T has exactly one inverse (which we denote as T^{-1}), and $T^{-1} : \mathbb{R}^n \to \mathbb{R}^n$ is the transformation induced by the matrix A^{-1}.

In other words, $(T_A)^{-1} = T_{A^{-1}}$.
T and T^{-1} 'reverse' or 'undo' the action of each other.

Activity: Reversing a reflection (U2-LO5)

Consider $T : \mathbb{R}^2 \to \mathbb{R}^2$ given by reflection across the x-axis. Find the matrix representing this transformation, and find the inverse.

T is given by matrix $A = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$

Since $A^2 = I_2$, A is its own inverse, so $T^{-1} = T$.

Theorem: Determinant and inverse of a 2×2 matrix

The 2×2 matrix

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

is invertible if and only if $ad - bc \neq 0$, in which case

$$A^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

We define the **determinant** of the matrix A to be the quantity $\det(A) = ad - bc$. Thus the matrix A is invertible if and only if its determinant is nonzero.

For now, $\det(A) = ad - bc$ is only the determinant of a 2×2 matrix. We will define determinants for larger square matrices later in this section.

Activity: Reading determinant checks in code (U2-LO3, U2-LO5, U2-LO7)

The following code computes determinants of two 2×2 matrices.

```
import numpy as np

A = np.array([
    [1., 1.],
    [0., 1.],
])

B = np.array([
    [1., 0.],
    [0., 0.],
])

detA = float(np.linalg.det(A))
detB = float(np.linalg.det(B))

detA, detB
```

Output:

```
(1.0, 0.0)
```

Activity: Reading determinant checks in code (U2-LO3, U2-LO5, U2-LO7) (continued)

1. Which matrix is invertible?
2. Which matrix forgets a nonzero input direction?
3. How does the determinant tell you?

The output says

$$\det(A) = 1, \qquad \det(B) = 0.$$

Matrix A is invertible because its determinant is nonzero. Matrix B is singular because its determinant is zero.

The matrix B forgets the nonzero input direction

$$\begin{bmatrix} 0 \\ 1 \end{bmatrix},$$

since

$$B \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}.$$

For a 2×2 matrix, a nonzero determinant means

the matrix is invertible. A zero determinant means the matrix is singular and forgets some nonzero input direction.

Activity: Checking invertibility and finding inverses (U2-LO5)

Find the inverse of each matrix, if possible.

$$A = \begin{bmatrix} -2 & -3 \\ 4 & 6 \end{bmatrix}$$

$$\det(A) = (-2)(6) - (-3)(4) = -12 - (-12) = 0,$$

so A is not invertible.

Activity: Checking invertibility and finding inverses (U2-LO5)

Find the inverse of each matrix, if possible.

$$B = \begin{bmatrix} 1 & 6 \\ 2 & 3 \end{bmatrix}$$

$$\det(B) = (1)(3) - (6)(2) = 3 - 12 = -9 \neq 0,$$

so

$$B^{-1} = \frac{1}{-9}\begin{bmatrix} 3 & -6 \\ -2 & 1 \end{bmatrix} = \begin{bmatrix} -1/3 & 2/3 \\ 2/9 & -1/9 \end{bmatrix}.$$

If A is an $n \times n$ matrix, then the linear system $A\mathbf{x} = \mathbf{b}$ is a system of n equations in n unknowns. Suppose A is nonsingular. How can we use A^{-1} to solve the system $A\mathbf{x} = \mathbf{b}$?

Activity: Solving a system with an inverse (U2-LO5)

Solve the system using the inverse of A.

Multiply both sides of $A\mathbf{x} = \mathbf{b}$ by A^{-1} on the left:

$$
\begin{aligned}
A\mathbf{x} &= \mathbf{b} \\
A^{-1}(A\mathbf{x}) &= A^{-1}\mathbf{b} \\
(A^{-1}A)\mathbf{x} &= A^{-1}\mathbf{b} \\
I_n\mathbf{x} &= A^{-1}\mathbf{b} \\
\mathbf{x} &= A^{-1}\mathbf{b}
\end{aligned}
$$

Therefore, when A is nonsingular, $\mathbf{x} = A^{-1}\mathbf{b}$ is the solution to $A\mathbf{x} = \mathbf{b}$.

Consequences:

- When A^{-1} exists, then $A\mathbf{x} = \mathbf{b}$ has a unique solution.
- If A is invertible/nonsingular, then the *only* solution to the homogeneous system $A\mathbf{x} = \mathbf{0}$ is $\mathbf{x} = \mathbf{0}$.
- Put differently, an invertible square matrix forgets no nonzero input direction.

Activity: One inverse, several right-hand sides (U2-LO5)

Suppose that

$$
A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 4 \\ 2 \end{bmatrix}, \quad \mathbf{c} = \begin{bmatrix} 3 \\ 2 \end{bmatrix}.
$$

Use this information to calculate the solution to the following systems:
$A\mathbf{x} = \mathbf{b}$

Since $\det(A) = (1)(4) - (2)(3) = -2 \neq 0$, the 2×2 inverse formula in Determinant and inverse of a 2×2 matrix gives

$$A^{-1} = -\frac{1}{2}\begin{bmatrix} 4 & -2 \\ -3 & 1 \end{bmatrix} = \begin{bmatrix} -2 & 1 \\ \frac{3}{2} & -\frac{1}{2} \end{bmatrix}.$$

Therefore,

$$\mathbf{x} = A^{-1}\mathbf{b} = \begin{bmatrix} -6 \\ 5 \end{bmatrix}$$

Check this solution by elimination in The Elimination Scratchpad

Activity: One inverse, several right-hand sides (U2-LO5)

Suppose that

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 4 \\ 2 \end{bmatrix}, \quad \mathbf{c} = \begin{bmatrix} 3 \\ 2 \end{bmatrix}.$$

Use this information to calculate the solution to the following systems:
$A\mathbf{x} = \mathbf{c}$

$$\mathbf{x} = A^{-1}\mathbf{c} = \begin{bmatrix} -4 \\ \frac{7}{2} \end{bmatrix}$$

Check this solution by elimination in The Elimination Scratchpad

Activity: One inverse, several right-hand sides (U2-LO5)

Suppose that

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 4 \\ 2 \end{bmatrix}, \quad \mathbf{c} = \begin{bmatrix} 3 \\ 2 \end{bmatrix}.$$

Use this information to calculate the solution to the following systems:

$A\mathbf{x} = \mathbf{0}$

$$\mathbf{x} = A^{-1}\mathbf{0} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

Check this solution by elimination in The Elimination Scratchpad

Activity: A coordinate map and invertibility (U2-LO5)

Consider the transformation $P : \mathbb{R}^2 \to \mathbb{R}^2$ defined by

$$P\left(\begin{bmatrix} a_1 \\ a_2 \end{bmatrix}\right) = \begin{bmatrix} a_1 \\ 0 \end{bmatrix}.$$

Is P a matrix transformation? If so, find the matrix A corresponding to P.

Yes. The map P preserves addition and scalar multiplication, so it is linear. By Every linear map has a matrix, its matrix has the images of the standard basis vectors as columns:

$$A = \begin{bmatrix} P(\mathbf{e}_1) & P(\mathbf{e}_2) \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}.$$

To verify:

$$P\left(\begin{bmatrix} a_1 \\ a_2 \end{bmatrix}\right) = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}\begin{bmatrix} a_1 \\ a_2 \end{bmatrix} = \begin{bmatrix} a_1 \\ 0 \end{bmatrix}$$

Activity: A coordinate map and invertibility (U2-LO5)

Consider the transformation $P : \mathbb{R}^2 \to \mathbb{R}^2$ defined by

Activity: A coordinate map and invertibility (U2-LO5) (continued)

$$P\left(\begin{bmatrix} a_1 \\ a_2 \end{bmatrix}\right) = \begin{bmatrix} a_1 \\ 0 \end{bmatrix}.$$

Is P an invertible transformation?

No, P is not invertible. Note that $\det(A) = 0$, and so A is not an invertible matrix, and so P cannot be an invertible transformation.

Consider vectors $\mathbf{c}_1, \ldots, \mathbf{c}_n$ in $\mathbb{R}^m$, and let A be the $m \times n$ matrix whose columns are the vectors $\mathbf{c}_1, \ldots, \mathbf{c}_n$. Recall that $\{\mathbf{c}_1, \ldots, \mathbf{c}_n\}$ are linearly independent precisely when the equation $A\mathbf{x} = \mathbf{0}$ has *only* the *trivial* solution (see Linear Independence and the uniqueness theorem, Uniqueness of linear-combination coefficients). By Spanning and solvability, $\text{span}\{\mathbf{c}_1, \ldots, \mathbf{c}_n\} = \mathbb{R}^m$ exactly when $A\mathbf{x} = \mathbf{b}$ has a solution for all $\mathbf{b}$ in $\mathbb{R}^m$.

Theorem: Invertibility Criteria

Let A be an $n \times n$ matrix. The following are equivalent:

1. A is nonsingular/invertible.
2. The columns of A are linearly independent.
3. The columns of A span $\mathbb{R}^n$.
4. The rows of A are linearly independent.
5. The rows of A span $\mathbb{R}^n$.
6. $A\mathbf{x} = \mathbf{0}$ has only the trivial solution.
7. A is row equivalent to I_n.

8. The linear system $A\mathbf{x} = \mathbf{b}$ has a unique solution for every $n \times 1$ matrix $\mathbf{b}$.

9. The rank of A is n.

Activity: Testing dependence using a matrix (U2-LO4)

Let $S = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$, where

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix} \quad \mathbf{v}_2 = \begin{bmatrix} 0 \\ -1 \\ 2 \end{bmatrix} \quad \mathbf{v}_3 = \begin{bmatrix} 3 \\ 0 \\ -5 \end{bmatrix}.$$

Is S linearly dependent?

Put the vectors in the columns of a matrix:

$$A = \begin{bmatrix} 1 & 0 & 3 \\ 2 & -1 & 0 \\ 2 & 2 & -5 \end{bmatrix}$$

Practice the homogeneous system in The Elimination Scratchpad Use c_1, c_2, c_3 for the coefficients in $c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + c_3\mathbf{v}_3 = \mathbf{0}$.
Following the Gaussian elimination algorithm, we scale each pivot to 1 and eliminate the entries below it:

$$\begin{bmatrix} 1 & 0 & 3 \\ 2 & -1 & 0 \\ 2 & 2 & -5 \end{bmatrix}$$

$$\xrightarrow[R_3 \to R_3 - 2R_1]{R_2 \to R_2 - 2R_1} \begin{bmatrix} 1 & 0 & 3 \\ 0 & -1 & -6 \\ 0 & 2 & -11 \end{bmatrix}$$

$$\xrightarrow{R_2 \to -R_2} \begin{bmatrix} 1 & 0 & 3 \\ 0 & 1 & 6 \\ 0 & 2 & -11 \end{bmatrix}$$

$$\xrightarrow{R_3 \to R_3 - 2R_2} \begin{bmatrix} 1 & 0 & 3 \\ 0 & 1 & 6 \\ 0 & 0 & -23 \end{bmatrix}$$

$$\xrightarrow{R_3 \to -\frac{1}{23}R_3} \begin{bmatrix} 1 & 0 & 3 \\ 0 & 1 & 6 \\ 0 & 0 & 1 \end{bmatrix}.$$

This echelon form has a pivot in every column. Therefore the equation $Ac = 0$ has only the trivial solution, so no nontrivial linear combination of $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$ gives zero. Thus S is linearly independent, not linearly dependent.

General Determinants

We generalize the determinant to $n \times n$ matrices, which in particular gives us a way to determine whether a matrix is invertible. We use the 2×2 determinant from Determinant and inverse of a 2×2 matrix to define the 3×3 determinant. Later, an $n \times n$ determinant will be built from $(n-1) \times (n-1)$ determinants.

Definition: Minors of a 3×3 matrix

Let A be a 3×3 matrix. The (i, j)-th minor of A, denoted

$$\text{minor}(A)_{ij},$$

is the determinant of the 2×2 matrix obtained from A by deleting row i and column j.

Activity: Computing matrix minors (U2-LO7)

Suppose

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 3 & 2 \\ 3 & 2 & 1 \end{bmatrix}.$$

Calculate $\text{minor}(A)_{11}$.

$$\text{minor}(A)_{11} = \det \begin{bmatrix} 3 & 2 \\ 2 & 1 \end{bmatrix} = 3(1) - 2(2) = -1.$$

Activity: Computing matrix minors (U2-LO7)

Suppose

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 3 & 2 \\ 3 & 2 & 1 \end{bmatrix}.$$

Calculate $\text{minor}(A)_{23}$.

$$\text{minor}(A)_{23} = \det \begin{bmatrix} 1 & 2 \\ 3 & 2 \end{bmatrix} = 1(2) - 2(3) = -4.$$

Definition: Cofactors of a 3×3 matrix

Let A be a 3×3 matrix. The (i, j)-cofactor of A, denoted

$$\text{cof}(A)_{ij},$$

is

$$\text{cof}(A)_{ij} = (-1)^{i+j} \, \text{minor}(A)_{ij}.$$

Activity: Computing cofactors (U2-LO7)

Define

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 3 & 2 \\ 3 & 2 & 1 \end{bmatrix}$$

Compute $\text{cof}(A)_{11}$.

$$\operatorname{cof}(A)_{11} = (-1)^{1+1} \operatorname{minor}(A)_{11} = (1)(-1) = -1.$$

Activity: Computing cofactors (U2-LO7)

Define

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 3 & 2 \\ 3 & 2 & 1 \end{bmatrix}$$

Compute $\operatorname{cof}(A)_{23}$.

$$\operatorname{cof}(A)_{23} = (-1)^{2+3} \operatorname{minor}(A)_{23} = (-1)(-4) = 4.$$

Definition: Determinant of a 3×3 matrix

Let $A = [a_{ij}]$ be a 3×3 matrix. The determinant of A, denoted $\det(A)$, is defined by expansion along the first row:

$$\det(A) = a_{11} \operatorname{cof}(A)_{11} + a_{12} \operatorname{cof}(A)_{12} + a_{13} \operatorname{cof}(A)_{13}.$$

This process is called cofactor expansion.

Fact

For a 3×3 matrix, cofactor expansion along any row or any column gives the same determinant.

For example, expanding along the second row means multiplying each entry in that row by its cofactor and adding:

$$\det(A) = a_{21}\,\mathrm{cof}(A)_{21} + a_{22}\,\mathrm{cof}(A)_{22} + a_{23}\,\mathrm{cof}(A)_{23}.$$

Expanding along the second column instead means multiplying each entry in that column by its cofactor and adding:

$$\det(A) = a_{12}\,\mathrm{cof}(A)_{12} + a_{22}\,\mathrm{cof}(A)_{22} + a_{32}\,\mathrm{cof}(A)_{32}.$$

Activity: Comparing row and column expansions (U2-LO7)

Let

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 3 & 2 \\ 3 & 2 & 1 \end{bmatrix}.$$

Find the determinant of A by expanding along row 1.

To calculate the determinant of A by expanding along row 1, we write

$$\det(A) = 1\,\mathrm{cof}(A)_{11} + 2\,\mathrm{cof}(A)_{12} + 3\,\mathrm{cof}(A)_{13}.$$

We begin by calculating the cofactors:

$$\begin{aligned}\operatorname{cof}(A)_{11} &= (-1)^{1+1} \det \begin{bmatrix} 3 & 2 \\ 2 & 1 \end{bmatrix} \\ &= (1)(3 \cdot 1 - 2 \cdot 2) = -1\end{aligned}$$

$$\begin{aligned}\operatorname{cof}(A)_{12} &= (-1)^{1+2} \det \begin{bmatrix} 4 & 2 \\ 3 & 1 \end{bmatrix} \\ &= (-1)(4 \cdot 1 - 2 \cdot 3) = 2\end{aligned}$$

$$\begin{aligned}\operatorname{cof}(A)_{13} &= (-1)^{1+3} \det \begin{bmatrix} 4 & 3 \\ 3 & 2 \end{bmatrix} \\ &= (1)(8 - 9) = -1\end{aligned}$$

Therefore

$$\begin{aligned}\det(A) &= 1(-1) + 2(2) + 3(-1) \\ &= -1 + 4 - 3 = 0.\end{aligned}$$

Activity: Comparing row and column expansions (U2-LO7)

Let

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 3 & 2 \\ 3 & 2 & 1 \end{bmatrix}.$$

Find the determinant of A by expanding along column 2.

To calculate the determinant of A by expanding along column 2, we write

$$\det(A) = 2\,\mathrm{cof}(A)_{12} + 3\,\mathrm{cof}(A)_{22} + 2\,\mathrm{cof}(A)_{32}$$

We calculate the cofactors:

$$\begin{aligned} \mathrm{cof}(A)_{12} &= (-1)^{1+2} \det \begin{bmatrix} 4 & 2 \\ 3 & 1 \end{bmatrix} \\ &= (-1)(4 \cdot 1 - 2 \cdot 3) = 2 \end{aligned}$$

$$\begin{aligned} \mathrm{cof}(A)_{22} &= (-1)^{2+2} \det \begin{bmatrix} 1 & 3 \\ 3 & 1 \end{bmatrix} \\ &= (1)(1 \cdot 1 - 3 \cdot 3) = -8 \end{aligned}$$

$$\begin{aligned}\operatorname{cof}(A)_{32} &= (-1)^{3+2} \det \begin{bmatrix} 1 & 3 \\ 4 & 2 \end{bmatrix} \\ &= (-1)(2-12) = 10.\end{aligned}$$

Therefore

$$\begin{aligned}\det(A) &= 2(2) + 3(-8) + 2(10) \\ &= 4 - 24 + 20 = 0.\end{aligned}$$

Note

To track the sign $(-1)^{i+j}$ when computing cofactors, it may be helpful to view the matrix as a checkerboard as below:

$$\begin{bmatrix} + & - & + \\ - & + & - \\ + & - & + \end{bmatrix}$$

The sign in the ith row and jth position with be the sign of $(-1)^{i+j}$.

Note

Since any row or column gives the same determinant, choose a row or column with many zeros when possible.

Activity: Choosing a determinant expansion (U2-LO7)

Compute the determinant of the following matrices:

$$A = \begin{bmatrix} 3 & -5 & 1 \\ 0 & 1 & 0 \\ -2 & -4 & 7 \end{bmatrix}$$

The second row of A has two zeros, and so expanding along this row will likely result in the simplest computation of the determinant. We write

$$\begin{aligned} \det(A) &= 0 \cdot \operatorname{cof}(A)_{21} + 1 \cdot \operatorname{cof}(A)_{22} + 0 \cdot \operatorname{cof}(A)_{23} \\ &= \operatorname{cof}(A)_{22} \\ &= (-1)^{2+2} \det \begin{bmatrix} 3 & 1 \\ -2 & 7 \end{bmatrix} \\ &= (1)(3 \cdot 7 - 1 \cdot (-2)) \\ &= 21 + 2 = 23. \end{aligned}$$

Activity: Choosing a determinant expansion (U2-LO7)

Compute the determinant of the following matrices:

$$B = \begin{bmatrix} 0 & \pi & 200 \\ 0 & 0.03 & 4 \\ 0 & 19 & -3584 \end{bmatrix}$$

If we expand the determinant of B along its first column, we find that

$$\begin{aligned} \det(B) &= 0 \cdot \operatorname{cof}(B)_{11} \\ &\quad + 0 \cdot \operatorname{cof}(B)_{21} \\ &\quad + 0 \cdot \operatorname{cof}(B)_{31} \\ &= 0. \end{aligned}$$

Note

Whenever a row or column of a matrix only has zeros as entries, expanding along that row shows that the determinant of the matrix is always zero, so no cofactor computations are required.

Now that 3×3 determinants have been defined, we can define larger determinants recursively. For a 4×4 matrix, each minor determinant is a 3×3 determinant. For a 5×5 matrix, each minor determinant is a 4×4 determinant, and so on.

Definition: Determinant of a general square matrix

Let $A = [a_{ij}]$ be an $n \times n$ matrix with $n \geq 4$. Assume determinants have already been defined for $(n-1) \times (n-1)$ matrices.
The (i, j)-th minor of A, denoted

$$\text{minor}(A)_{ij},$$

is the determinant of the $(n-1) \times (n-1)$ matrix obtained from A by deleting row i and column j.
The (i, j)-cofactor of A is

$$\text{cof}(A)_{ij} = (-1)^{i+j} \, \text{minor}(A)_{ij}.$$

The determinant of A is defined by expansion along the first row:

$$\det(A) = \sum_{j=1}^{n} a_{1j} \, \text{cof}(A)_{1j}.$$

Fact

For an $n \times n$ matrix, cofactor expansion along any row or any column gives the same determinant:

$$\det(A) = \sum_{j=1}^{n} a_{ij} \, \text{cof}(A)_{ij}$$

for any fixed row i, and

$$\det(A) = \sum_{i=1}^{n} a_{ij} \, \text{cof}(A)_{ij}$$

for any fixed column j.

Example: A four-by-four determinant (U2-LO7)

Calculate the determinant of the matrix

$$A = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 5 & 4 & 2 & 3 \\ 1 & 3 & 4 & 5 \\ 3 & 4 & 3 & 2 \end{bmatrix}$$

Expand along column 3:

$$\det(A) = 3\,\mathrm{cof}(A)_{13} + 2\,\mathrm{cof}(A)_{23} + 4\,\mathrm{cof}(A)_{33} + 3\,\mathrm{cof}(A)_{43}.$$

The signs in column 3 are

$$+,\ -,\ +,\ -.$$

So

$$\begin{aligned}\det(A) = 3\det\begin{bmatrix}5 & 4 & 3\\ 1 & 3 & 5\\ 3 & 4 & 2\end{bmatrix}\\ -2\det\begin{bmatrix}1 & 2 & 4\\ 1 & 3 & 5\\ 3 & 4 & 2\end{bmatrix}\\ +4\det\begin{bmatrix}1 & 2 & 4\\ 5 & 4 & 3\\ 3 & 4 & 2\end{bmatrix}\\ -3\det\begin{bmatrix}1 & 2 & 4\\ 5 & 4 & 3\\ 1 & 3 & 5\end{bmatrix}.\end{aligned}$$

Using the 3×3 determinant method,

$$\det\begin{bmatrix}5 & 4 & 3\\ 1 & 3 & 5\\ 3 & 4 & 2\end{bmatrix} = -33,$$

$$\det\begin{bmatrix}1 & 2 & 4\\ 1 & 3 & 5\\ 3 & 4 & 2\end{bmatrix} = -8,$$

$$\det\begin{bmatrix} 1 & 2 & 4 \\ 5 & 4 & 3 \\ 3 & 4 & 2 \end{bmatrix} = 26,$$

and

$$\det\begin{bmatrix} 1 & 2 & 4 \\ 5 & 4 & 3 \\ 1 & 3 & 5 \end{bmatrix} = 11.$$

Therefore

$$\begin{aligned} \det(A) &= 3(-33) - 2(-8) + 4(26) - 3(11) \\ &= -99 + 16 + 104 - 33 \\ &= -12. \end{aligned}$$

Activity: The determinant of a triangular matrix (U2-LO7)

Compute $\det(A)$, where

$$A = \begin{bmatrix} 3 & 2 & 3 \\ 0 & 4 & 2 \\ 0 & 0 & 2 \end{bmatrix}.$$

Expanding in cofactors along column 1 (because most of the entries of that column are zeros), we find

$$\begin{aligned}\det(A) &= 3\,\mathrm{cof}(A)_{11} + 0\,\mathrm{cof}(A)_{21} \\ &\quad + 0\,\mathrm{cof}(A)_{31} \\ &= 3(-1)^{1+1}\det\begin{bmatrix}4 & 2\\ 0 & 2\end{bmatrix} \\ &= 3((4)(2)-(2)(0)) \\ &= 24.\end{aligned}$$

Recall Main diagonal.

Definition: Triangular and Diagonal Matrices

An $n \times n$ matrix $A = [a_{ij}]$ is called upper triangular if $a_{ij} = 0$ for $i > j$, and lower triangular if $a_{ij} = 0$ for $i < j$. A matrix is diagonal if it is both upper triangular and lower triangular, i.e., if $a_{ij} = 0$ for $i \neq j$. We denote the $n \times n$ diagonal matrix with entries $\lambda_1, \ldots, \lambda_n$ on the diagonal by $\mathrm{diag}(\lambda_1, \ldots, \lambda_n)$.

Theorem

If $A = [a_{ij}]$ is a upper or lower triangular matrix, then $\det(A) = a_{11} \cdots a_{nn}$. That is, the determinant of a triangular matrix is the product of entries in the *main diagonal*.

Why is this true?

A generalization of the argument used to calculate The determinant of a triangular matrix (U2-LO7), i.e., expanding the determinant in cofactors along the first column iteratively.

Theorem

If a row or column of a matrix A consists entirely of zeros, then $\det(A) = 0$.

Why is this true?

A generalization of the argument used to calculate the determinant in the activity.

Theorem: Row or column swaps reverse the determinant's sign

If a matrix B is obtained from a matrix A by swapping two rows or two columns, then $\det(B) = -\det(A)$.

Activity: Row swaps and determinants (U2-LO7)

Check directly that Row or column swaps reverse the determinant's sign is true when A and B are 2×2 matrices.

Why is this true?

If

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

and

Activity: Row swaps and determinants (U2-LO7) (continued)

$$B = \begin{bmatrix} c & d \\ a & b \end{bmatrix},$$

then $\det(A) = ad - bc$ and $\det(B) = cb - ad = -(ad - bc) = -\det(A)$.

Theorem

If two rows (or columns) of A are equal to one another, then $\det(A) = 0$.

Why is this true?

Let B be obtained by swapping the two identical rows (or columns) of A. Then $A = B$, and Row or column swaps reverse the determinant's sign implies that $\det(A) = -\det(B) = -\det(A)$. Therefore $2\det(A) = 0$, and thus that $\det(A) = 0$.

Theorem

If B is obtained from A by multiplying each entry in a row or column of A by a quantity k, then $\det(B) = k\det(A)$.

Why is this true?

Suppose that B is obtained from A by multiplying each entry of row i by k. Consider a cofactor expansion along row i of B. Then

$$\begin{aligned} \det(B) &= ka_{i1}\text{cof}(B)_{i1} + ka_{i2}\text{cof}(B)_{i2} + \cdots + ka_{in}\text{cof}(B)_{in} \\ &= k[a_{i1}\text{cof}(A)_{i1} + \cdots + a_{in}\text{cof}(A)_{in}] \\ &= k\det(A). \end{aligned}$$

Fact

If A is a $n \times n$ matrix, then $\det(kA) = k^n \det(A)$.

Theorem: Row replacement preserves the determinant

If B is obtained by adding k times the ith row of A to the jth row, for

$i \neq j$, which is a row operation which we denote by $R_j \to R_j + kR_i$, then $\det(B) = \det(A)$.

We can check Row replacement preserves the determinant in the 2×2 case. If

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

and

$$B = \begin{bmatrix} a & b \\ c + ka & d + kb \end{bmatrix},$$

then

$$\det(A) = ad - bc$$

and

$$\begin{aligned} \det(B) &= a(d + kb) - b(c + ka) \\ &= ad + kab - bc - kab \\ &= ad - bc = \det(A). \end{aligned}$$

Theorem: How row operations affect the determinant

Here is a summary of how the elementary row operations, when applied to a matrix, affect the determinant:

- $R_i \to kR_i$ multiplies the determinant by k.
- $R_i \leftrightarrow R_j$ multiplies the determinant by -1.
- $R_j \to R_j + kR_i$ leaves the determinant unchanged.

Activity: Tracking determinants through row operations (U2-LO7)

Suppose a matrix B is obtained from a matrix A by applying the following row operations:

- $R_i \leftrightarrow R_j$
- $R_4 \to 4R_4$
- $R_3 \to R_3 + 3R_2$

If $\det(A) = 8$, what is the determinant of B.

- $R_i \leftrightarrow R_j$ multiplies the determinant by -1.
- $R_4 \to 4R_4$ multiplies the determinant by 4.
- $R_3 \to R_3 + 3R_2$ does not change the determinant.

Therefore $\det(B) = (-1)(4)\det(A) = -32$.

Activity: Evaluating a three-by-three determinant (U2-LO7)

Compute the determinant of the matrix

$$B = \begin{bmatrix} 1 & 2 & 1 \\ -1 & 3 & 3 \\ 0 & 15 & 12 \end{bmatrix}.$$

We use the elementary row operations to make the matrix B triangular:

$$R_2 \to R_2 + R_1$$

$$\begin{bmatrix} 1 & 2 & 1 \\ 0 & 5 & 4 \\ 0 & 15 & 12 \end{bmatrix}$$

$$R_3 \to R_3 - 3R_2$$

$$\begin{bmatrix} 1 & 2 & 1 \\ 0 & 5 & 4 \\ 0 & 0 & 0 \end{bmatrix}$$

None of the row operations we have performed change the determinant of the matrix, and the final matrix has determinant zero, so $\det(B) = 0$.

Theorem: Determinant of a product

If A and B are $n \times n$ matrices, then $\det(AB) = \det(A)\det(B)$.

Activity: Determinants of matrices and their products (U2-LO7)

Let

$$A = \begin{bmatrix} 3 & 1 \\ -2 & 7 \end{bmatrix} \quad B = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix} \quad C = \begin{bmatrix} 1 & 4 \\ 0 & 2 \end{bmatrix}.$$

Compute the determinant of A, B, and C.

We use the formula for 2×2 matrices: First, that

$$\det(A) = (3)(7) - (1)(-2) = 23,$$

then that

$$\det(B) = (1)(4) - (2)(2) = 0,$$

and that

$$\det(C) = (1)(2) - (4)(0) = 2.$$

Activity: Determinants of matrices and their products (U2-LO7)

Let

$$A = \begin{bmatrix} 3 & 1 \\ -2 & 7 \end{bmatrix} \quad B = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix} \quad C = \begin{bmatrix} 1 & 4 \\ 0 & 2 \end{bmatrix}.$$

Compute the determinant of AC and BC.

Using Determinant of a product, we calculate that

$$\det(AC) = \det(A)\det(C) = (23)(2) = 46,$$

and that

$$\det(BC) = \det(B)\det(C) = (0)(2) = 0.$$

Alternatively, we can calculate the products of the matrices, i.e., that

$$AC = \begin{bmatrix} 3 & 14 \\ -2 & 6 \end{bmatrix}$$

and that

$$BC = \begin{bmatrix} 1 & 8 \\ 2 & 16 \end{bmatrix},$$

and calculate the determinants directly.

Theorem

If A is a $n \times n$ matrix, then A is *nonsingular* if and only if $\det(A) \neq 0$. In this case $\det(A^{-1}) = 1/\det(A)$.

Why is this true?

We prove only one direction of the theorem, that if A is nonsingular then $\det(A) \neq 0$ and $\det(A^{-1}) = 1/\det(A)$. Suppose that A is non-singular. Then $AA^{-1} = I$. Taking determinants of both sides, and using that I is diagonal, so that $\det(I) = 1$, we see that

$$1 = \det(I) = \det(AA^{-1}) = \det(A)\det(A^{-1}).$$

Thus $\det(A) \neq 0$ (since otherwise the right hand side of this equation would be zero), and rearranging gives $\det(A^{-1}) = 1/\det(A)$.

Theorem

If A is a square matrix, then $\det(A) = \det(A^T)$.

Definition: Orthogonal Matrices

A square matrix A is orthogonal if $A^T = A^{-1}$.

Activity: Determinants of orthogonal matrices (U2-LO7)

If A is an orthogonal matrix, what are the possible values of $\det(A)$?

If A is orthogonal, then applying the determinant to both sides of the equation $I = AA^T$ gives

$$\begin{aligned} 1 &= \det(I) = \det(AA^T) \\ &= \det(A)\det(A^T) = \det(A)^2. \end{aligned}$$

Taking square roots gives that $\det(A) = \pm 1$.